RecursiveCharacterTextSplitter

Do Smaller Chunks Lose Context?

Do Larger Chunks Dilute Relevance?

Chunk Size Trade-offs, Mitigations & Best Practices

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LangChain • RAG Pipelines • Retrieval Quality • Chunk Optimization

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1. Introduction — The Core Tension

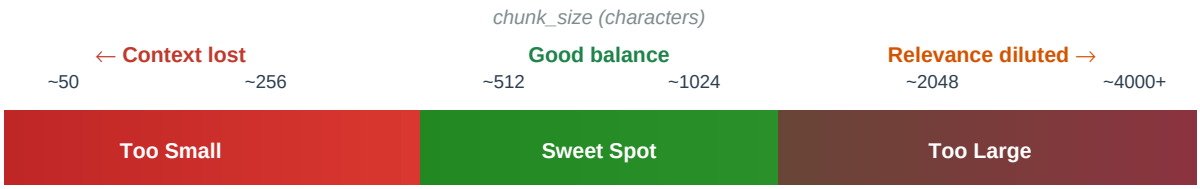
When building RAG (Retrieval-Augmented Generation) pipelines with **RecursiveCharacterTextSplitter**, the two most important parameters — chunk_size and chunk_overlap — control a fundamental tension that directly determines answer quality:

Too small → chunks are precise to retrieve but lack the surrounding sentences the LLM needs to understand and use them. **Too large** → chunks carry full context but their embeddings are diluted across multiple topics, making retrieval less accurate.

> ★ Neither extreme is correct. The goal is to find the chunk size where a single chunk contains exactly one coherent idea — no more, no less — with enough surrounding context for the LLM to interpret it correctly.

2. The Chunk Size Spectrum

Think of chunk size as a dial with two danger zones at either end and a 'sweet spot' in the middle that depends on your document type and query style.



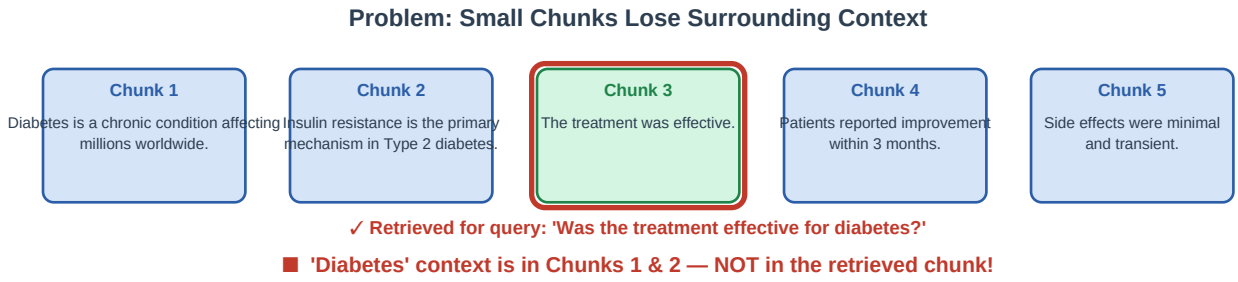
Rule of thumb: chunk_overlap = 10–20% of chunk_size • Start at 512, tune with real queries

Figure 1 — chunk_size spectrum from too-small to too-large (characters)

> ■ The sweet spot is not fixed — it shifts based on your embedding model's token limit, your document's average sentence length, and the nature of your queries. Always tune empirically.

3. Problem 1 — Small Chunks Lose Context

When chunk_size is very small (e.g. 50–150 characters), each chunk holds only one or two sentences. While this maximises retrieval precision — the retrieved chunk closely matches the query — it strips away the surrounding sentences that give those words their meaning.



The LLM receives Chunk 3 alone and cannot answer 'effective for diabetes?' — the disease name is missing.

Figure 2 — Query matches Chunk 3, but 'diabetes' context lives in Chunks 1 & 2

3.1 Why this matters

Consider the query *"Was the treatment effective for diabetes?"*. The retriever finds Chunk 3 — *"The treatment was effective."* — because it is the closest semantic match. But the LLM receives only that one sentence. It has no knowledge that the treatment was for diabetes, who the patients were, or what the treatment involved. The answer will be vague or wrong.

Symptoms of chunks that are too small:

- LLM answers are technically correct but miss the 'why' or 'who'
- Retrieved chunks seem relevant on the surface but lack actionable detail
- High retrieval score but low answer quality — a common confusion
- LLM frequently says 'I don't have enough information' even when the document has it

3.2 Code example — seeing the problem

```
from langchain_text_splitters import RecursiveCharacterTextSplitter

text = """
Diabetes is a chronic condition affecting millions worldwide.
Insulin resistance is the primary mechanism in Type 2 diabetes.
The treatment was effective.
Patients reported improvement within 3 months.
Side effects were minimal and transient.
"""

# TOO SMALL — each sentence becomes its own chunk
small_splitter = RecursiveCharacterTextSplitter(
    chunk_size=80,
    chunk_overlap=0,
)
chunks = small_splitter.split_text(text.strip())
```

```
for i, ch in enumerate(chunks):
    print(f"Chunk {i+1} ({len(ch):3d} chars): {ch}")

# Output:
# Chunk 1 ( 58 chars): Diabetes is a chronic condition affecting millions worldwide.
# Chunk 2 ( 66 chars): Insulin resistance is the primary mechanism in Type 2 diabetes.
# Chunk 3 ( 31 chars): The treatment was effective.        ← retrieved for query
# Chunk 4 ( 57 chars): Patients reported improvement within 3 months.
# Chunk 5 ( 42 chars): Side effects were minimal and transient.
#
# Query: "Was the treatment effective for diabetes?"
# Retrieved: Chunk 3 — contains NO mention of diabetes!
```

4. Problem 2 — Large Chunks Dilute Relevance

When chunk_size is very large (e.g. 3000–5000 characters), each chunk spans multiple paragraphs or sections covering different topics. The embedding model must represent *all* of those topics in a single vector — producing a diluted, averaged signal that matches no single query strongly.

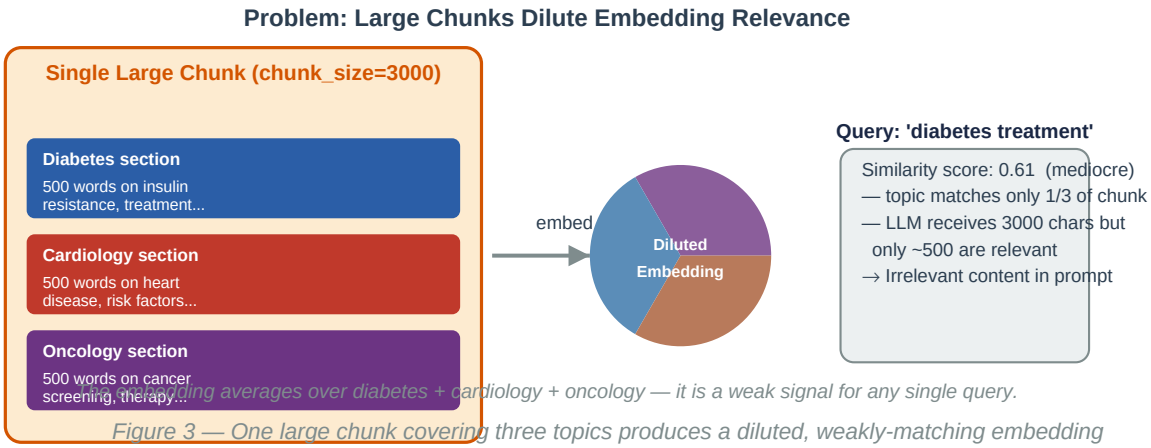


Figure 3 — One large chunk covering three topics produces a diluted, weakly-matching embedding

4.1 Why this matters

Consider a chunk of 3000 characters covering diabetes, cardiology, and oncology in roughly equal parts. When a user asks about diabetes treatment, the chunk's embedding has only ~33% diabetes signal. A smaller, diabetes-only chunk of 800 characters would have ~100% diabetes signal and rank much higher.

Symptoms of chunks that are too large:

- Retrieved chunks contain the right answer buried inside irrelevant paragraphs
- LLM receives 2000+ tokens but only 200 are relevant — wasting context budget
- Retrieval scores are consistently mediocre (0.55–0.65) across all queries
- MMR retrieval keeps returning the same few giant chunks for every query
- Chunk exceeds embedding model's token limit — text gets silently truncated

4.2 Code example — seeing the problem

```
from langchain_text_splitters import RecursiveCharacterTextSplitter
from langchain_openai import OpenAIEmbeddings
import numpy as np

embeddings = OpenAIEmbeddings(model="text-embedding-3-small")

diabetes_section    = "Diabetes treatment involves insulin... " * 20   # ~500 chars
cardiology_section  = "Cardiovascular disease risk factors... " * 20
oncology_section    = "Cancer screening protocols include... " * 20

# TOO LARGE — all three topics in one chunk
large_chunk  = diabetes_section + cardiology_section + oncology_section  # ~1500 chars
small_chunk  = diabetes_section   # diabetes only, ~500 chars

query = "What is the treatment for diabetes?"
q_emb         = embeddings.embed_query(query)
```

```
large_emb     = embeddings.embed_documents([large_chunk])[0]
small_emb     = embeddings.embed_documents([small_chunk])[0]

def cosine(a, b):
    a, b = np.array(a), np.array(b)
    return float(np.dot(a,b) / (np.linalg.norm(a) * np.linalg.norm(b)))

print(f"Large chunk similarity: {cosine(q_emb, large_emb):.3f}")  # ~0.61 (diluted)
print(f"Small chunk similarity: {cosine(q_emb, small_emb):.3f}")  # ~0.84 (focused)
# The focused small chunk ranks MUCH higher for the diabetes query.
```

5. The Trade-off Table

The table below summarises every dimension of the small-vs-large trade-off in one place.

Concern	Smaller Chunks	Larger Chunks
Retrieval precision	Higher — narrow, focused results	Lower — broad context returned
Context coherence	Lower — may cut mid-sentence/idea	Higher — full paragraphs preserved
Embedding quality	Sharp, focused signal	Diluted, averaged over many topics
LLM context quality	Misses surrounding context	Contains irrelevant content
Embedding cost	Higher (more chunks to embed)	Lower (fewer chunks)
LLM token usage	Lower per query	Higher per query
Risk	Answer present but context stripped	Low similarity score, poor ranking

Table 1 — Chunk size trade-off comparison across all key dimensions

6. Mitigation 1 — chunk_overlap

chunk_overlap is the most direct fix for lost boundary context. It repeats the last N characters of the previous chunk at the beginning of the next one, ensuring that sentences or ideas straddling a boundary appear in *both* adjacent chunks.

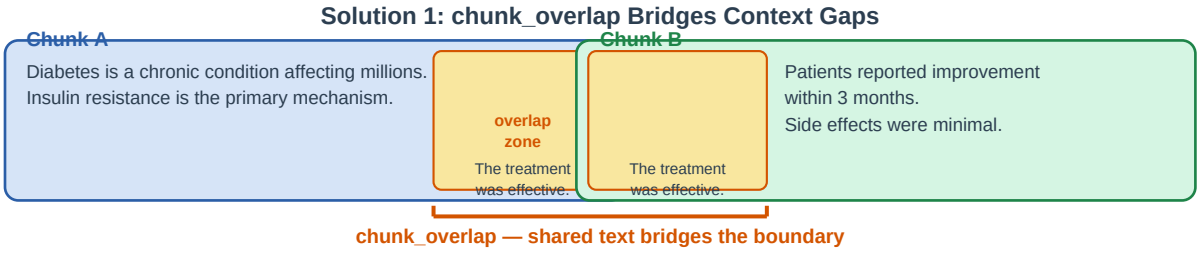


Figure 4 — chunk_overlap copies the end of Chunk A to the start of Chunk B

6.1 Overlap code example

```
from langchain_text_splitters import RecursiveCharacterTextSplitter

text = (
    "Diabetes is a chronic condition affecting millions worldwide. "
    "Insulin resistance is the primary mechanism in Type 2 diabetes. "
    "The treatment was effective. "
    "Patients reported improvement within 3 months. "
    "Side effects were minimal and transient."
)

splitter = RecursiveCharacterTextSplitter(
```

```
    chunk_size=120,
    chunk_overlap=40,   # ~33% overlap — last 40 chars repeated in next chunk
)
chunks = splitter.split_text(text)
for i, ch in enumerate(chunks):
    print(f"Chunk {i+1} ({len(ch)} chars): {ch}")

# Chunk 1: "...Insulin resistance is the primary mechanism in Type 2 diabetes."
# Chunk 2: "...Type 2 diabetes. The treatment was effective. Patients reported..."
#                    ^^^^^^^^^^^^^^^^^^^^^^^^
#           overlap carries 'diabetes' context into the next chunk!
```

■ chunk_overlap must always be strictly less than chunk_size. Setting overlap >= chunk_size causes a ValueError. Recommended range: overlap = 10–20% of chunk_size.

7. Mitigation 2 — Parent-Child Chunking

Parent-child chunking solves both problems simultaneously: index *small* chunks (precise retrieval) but fetch the *large* parent chunk (full context) to pass to the LLM. LangChain implements this with ParentDocumentRetriever.

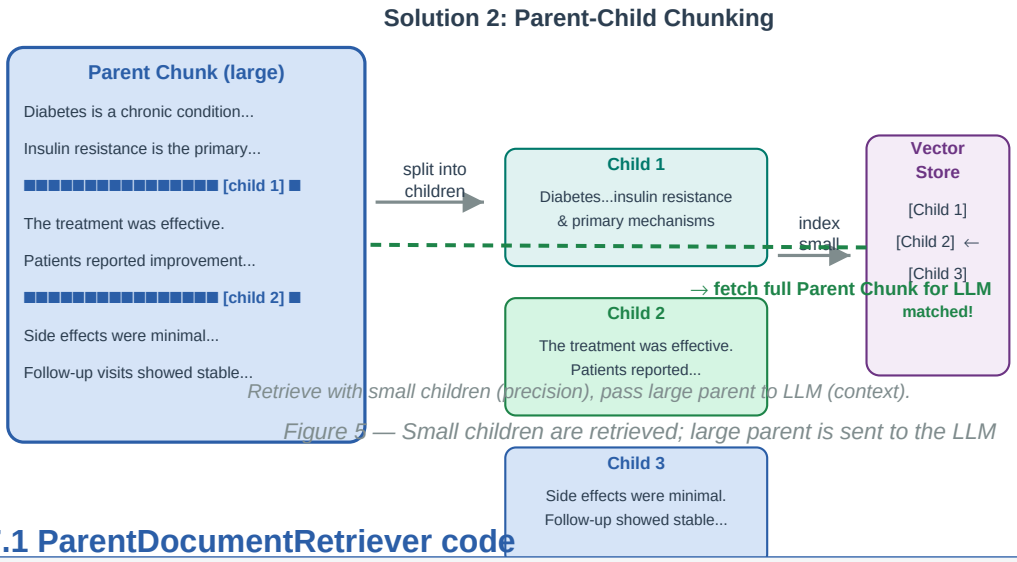


Figure 5 — Small children are retrieved; large parent is sent to the LLM

7.1 ParentDocumentRetriever code

```
from langchain.retrievers import ParentDocumentRetriever
from langchain.storage import InMemoryStore
from langchain_community.vectorstores import FAISS
from langchain_openai import OpenAIEmbeddings
from langchain_text_splitters import RecursiveCharacterTextSplitter
from langchain_community.document_loaders import TextLoader

# Load document
loader = TextLoader("medical_handbook.txt")
docs   = loader.load()

# Child splitter — small, precise chunks for indexing
child_splitter = RecursiveCharacterTextSplitter(
    chunk_size=200,
    chunk_overlap=20,
)

# Parent splitter — large chunks for LLM context
parent_splitter = RecursiveCharacterTextSplitter(
    chunk_size=1500,
    chunk_overlap=100,
)

# Storage: vector store for child embeddings, doc store for parent text
embeddings   = OpenAIEmbeddings(model="text-embedding-3-small")
vector_store = FAISS.from_documents([], embeddings)  # empty initially
doc_store    = InMemoryStore()

retriever = ParentDocumentRetriever(
    vectorstore=vector_store,
    docstore=doc_store,
    child_splitter=child_splitter,
    parent_splitter=parent_splitter,
```

```
)

# Index — splits into parents, then children; stores both
retriever.add_documents(docs)

# Query — retrieves children, returns parents
results = retriever.invoke("Was the treatment effective for diabetes?")
for doc in results:
    print(f"[{len(doc.page_content)} chars] {doc.page_content[:200]}...")
# Each result is a full PARENT chunk (~1500 chars) — rich context for the LLM.
```

8. Mitigation 3 — Contextual Retrieval

Contextual Retrieval (introduced by Anthropic) prepends a short, AI-generated summary of the chunk's document-level context to *every chunk* before embedding. Even a tiny 50-character chunk then carries enough context to rank correctly for complex queries.

The idea in one sentence: instead of embedding *"The treatment was effective."*, embed *"[Context: Section on Type 2 diabetes treatment outcomes in a 2024 clinical trial] The treatment was effective."*

8.1 Contextual Retrieval code

```
from anthropic import Anthropic
from langchain_text_splitters import RecursiveCharacterTextSplitter
from langchain_openai import OpenAIEmbeddings
from langchain_community.vectorstores import FAISS
from langchain.schema import Document

client     = Anthropic()
embeddings = OpenAIEmbeddings(model="text-embedding-3-small")

def generate_context(document: str, chunk: str) -> str:
    """Ask Claude to produce a 1-2 sentence context summary for the chunk."""
    response = client.messages.create(
        model="claude-3-haiku-20240307",
        max_tokens=120,
        messages=[{
            "role": "user",
            "content": (
                f"<document>{document[:4000]}</document>\n\n"
                f"Here is the chunk to situate:\n<chunk>{chunk}</chunk>\n\n"
                "Reply with 1-2 sentences that describe where this chunk fits in the document "
                "and what it is about. Be concise."
            ),
        }]
    )
    return response.content[0].text.strip()

def contextual_chunk_and_index(full_text: str, chunk_size=512, chunk_overlap=64):
    splitter = RecursiveCharacterTextSplitter(
        chunk_size=chunk_size,
        chunk_overlap=chunk_overlap,
    )
    raw_chunks = splitter.split_text(full_text)
    enriched_docs = []
    for i, chunk in enumerate(raw_chunks):
        context   = generate_context(full_text, chunk)
        enriched  = f"{context}\n\n{chunk}"   # prepend context
        enriched_docs.append(Document(
            page_content=enriched,
            metadata={"chunk_id": i, "original": chunk},
        ))
    return FAISS.from_documents(enriched_docs, embeddings)

# Usage
with open("research_paper.txt") as f:
```

```
    text = f.read()

store    = contextual_chunk_and_index(text)
results  = store.similarity_search("diabetes treatment outcomes", k=3)
for doc in results:
    print(doc.page_content[:300])
```

■ Contextual Retrieval adds one LLM call per chunk during indexing. Use a fast, cheap model (e.g. claude-haiku or gpt-4o-mini) for context generation to keep costs low.

9. Python Code Examples

9.1 Diagnosing your chunk size with similarity scores

```
from langchain_text_splitters import RecursiveCharacterTextSplitter
from langchain_openai import OpenAIEmbeddings
import numpy as np

def diagnose_chunk_size(
    text: str,
    sample_query: str,
    sizes: list[int] = [128, 256, 512, 1024, 2048],
):
    """
    For each chunk_size, embed all chunks and report the MAX cosine
    similarity against the sample_query. Higher max = better precision.
    """
    emb   = OpenAIEmbeddings(model="text-embedding-3-small")
    q_vec = np.array(emb.embed_query(sample_query))

    print(f"Query: '{sample_query}'\n{'■'*60}")
    print(f"{'chunk_size':>12} {'chunks':>8} {'avg_len':>9} {'max_sim':>9}")
    print('■'*60)

    for size in sizes:
        splitter = RecursiveCharacterTextSplitter(
            chunk_size=size, chunk_overlap=size//10
        )
        chunks = splitter.split_text(text)
        vecs   = np.array(emb.embed_documents(chunks))
        sims   = vecs @ q_vec / (np.linalg.norm(vecs, axis=1) * np.linalg.norm(q_vec))
        avg_l  = int(np.mean([len(c) for c in chunks]))
        print(f"{size:>12} {len(chunks):>8} {avg_l:>9} {sims.max():>9.3f}")

# Usage
with open("my_document.txt") as f:
    doc_text = f.read()

diagnose_chunk_size(doc_text, "What are the diabetes treatment outcomes?")

# Example output:
# chunk_size    chunks   avg_len   max_sim
# ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
#        128       312       115     0.821
#        256       162       234     0.847   ← good
#        512        84       481     0.863   ← best here
#       1024        44       940     0.831
#       2048        23      1820     0.794   ← dilution begins
```

9.2 Adaptive overlap based on chunk_size

```
from langchain_text_splitters import RecursiveCharacterTextSplitter

def make_splitter(
    chunk_size: int,
    overlap_ratio: float = 0.15,   # 15% overlap by default
```

```
) -> RecursiveCharacterTextSplitter:
    """Create a splitter with overlap automatically set to overlap_ratio * chunk_size."""
    overlap = int(chunk_size * overlap_ratio)
    return RecursiveCharacterTextSplitter(
        chunk_size=chunk_size,
        chunk_overlap=overlap,
        length_function=len,
        add_start_index=True,   # track character positions
    )

# Examples
splitter_small  = make_splitter(256)   # overlap=38
splitter_medium = make_splitter(512)   # overlap=76  (recommended start)
splitter_large  = make_splitter(1024)  # overlap=153

docs   = splitter_medium.split_documents(my_langchain_documents)
chunks = splitter_medium.split_text(my_raw_text)
```

10. Recommended Settings by Use Case

Use Case	chunk_size	chunk_overlap	Why
Dense keyword search	`256`	`32`	Maximum precision, short factual answers
Balanced RAG (default)	`512`	`64`	Best starting point for most pipelines
Contextual Q&A	`1024`	`128`	Preserves multi-sentence reasoning chains
Summarisation tasks	`2048`	`200`	Needs broad context per chunk
Full-page / big windows	`4000`	`400`	GPT-4 / Claude 3 with large context windows

Table 2 — Recommended chunk_size / chunk_overlap pairs by use case

■ These are starting points, not rules. Always run 10–20 representative queries against your specific data and adjust chunk_size until retrieved chunks consistently contain the full answer plus enough surrounding context.

11. Choosing the Right chunk_size: Decision Guide

Symptom: My answers are vague / miss 'who' and 'why'	**Fix:** Chunks are too small. Increase chunk_size or add chunk_overlap.
Symptom: LLM says 'not enough information' despite document having the answer	**Fix:** Chunks are too small — key context is in an adjacent chunk. Increase chunk_overlap or use ParentDocumentRetriever.
Symptom: Retrieval scores are consistently mediocre (0.55–0.70)	**Fix:** Chunks are too large — embedding is diluted. Decrease chunk_size.
Symptom: LLM receives irrelevant paragraphs alongside the answer	**Fix:** Chunks are too large. Decrease chunk_size.
Symptom: Correct chunks retrieved but answer still weak	**Fix:** Try Contextual Retrieval — prepend document-level context to each chunk before embedding.
Symptom: Same chunks retrieved for very different queries	**Fix:** Enable MMR retrieval (search_type='mmr') to diversify results.
Symptom: Token limit errors from embedding model	**Fix:** chunk_size exceeds model's token limit. text-embedding-3-small supports 8191 tokens (~6000 chars).

12. Mitigation Strategy Summary

Mitigation	Fixes	Cost	LangChain Class
chunk_overlap	Lost context at boundaries	None	`RecursiveCharacterTextSplitter(chunk_overlap=N)`
Parent-child chunking	Context vs. precision trade-off	Low	`ParentDocumentRetriever`
Contextual Retrieval	All chunk context loss	Medium	`Custom (Anthropic pattern)`
Sentence-window RAG	Narrow sentence chunks	Low	`LlamaIndex SentenceWindowNodeParser`
MMR retrieval	Duplicate/redundant chunks	None	`as_retriever(search_type='mmr')`

Table 3 — Mitigation strategies with what they fix, cost, and LangChain class

13. Common Pitfalls & Checklist

■ ***Setting chunk_overlap=0***	Guarantees information loss at every boundary. Always use overlap, even if small (5–10%).
■ ***Using chunk_size without testing on real queries***	The 'right' chunk_size is dataset-specific. What works for Wikipedia articles may fail on legal contracts or code files.
■ ***Ignoring the embedding model's token limit***	text-embedding-ada-002 supports 8191 tokens. A chunk_size of 8000 characters may truncate silently. Use a tokeniser to verify.
■ ***Assuming bigger overlap always helps***	chunk_overlap > 30% of chunk_size wastes embedding budget on duplicate content and can confuse the retriever with near-identical chunks.
■ ***Never inspecting retrieved chunks***	Always print the top-3 retrieved chunks for your 10 most important queries before going to production. Surprises are common.

13.1 Quick Checklist

- ■ chunk_overlap is 10–20% of chunk_size (never 0, never ≥ chunk_size)
- ■ Tested at least 3 different chunk_size values with real queries
- ■ Checked embedding model token limit — no silent truncation
- ■ Inspected top-3 retrieved chunks for 10+ representative queries
- ■ Used ParentDocumentRetriever if context loss persists after tuning
- ■ Considered Contextual Retrieval for complex, multi-topic documents
- ■ Enabled MMR retrieval to avoid duplicate chunk retrieval
- ■ add_start_index=True set for citation/debugging support

14. References

[1] LangChain Docs — RecursiveCharacterTextSplitter https://python.langchain.com/docs/how_to/recursive_text_splitter/

[2] LangChain Docs — ParentDocumentRetriever https://python.langchain.com/docs/how_to/parent_document_retriever/

[3] Anthropic — Contextual Retrieval https://www.anthropic.com/news/contextual-retrieval

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[5] OpenAI — Embeddings documentation (token limits) https://platform.openai.com/docs/guides/embeddings

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